Biochemistry Cume

December 1, 2006

A. (10 points) On average, biological membranes contain about 50% protein and 50% lipid. The structure and function of biological membranes is most often described in terms of the Singer-Nicholson Fluid-Mosaic Model first described in the early 1970's. Describe the general features of the Fluid-Mosaic Model.

B. (20 points) Let us assume that you are interested in lipid motions in a biological membrane. Describe an experimental method which could be used to determine the rate of motion of a lipid molecule in the plane of the membrane.

C. (20 points) Describe an experimental method which could be used to determine the rate of "flip-flop" of a lipid molecule from one half of the lipid bilayer to the other.

D. (20 points) Describe an experimental approach which could be used to examine the movement of a specific protein in the plane of the membrane.

E. (15 points) What is meant by the "melting temperature" of a membrane?

F. (15 points) What are lipid rafts and how are they relevant to the general theme of this exam?